

SalientGS: Supplementary Material

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1 Overview

This supplementary material accompanies the main paper “*SalientGS: Unified SfM-to-3DGS with Importance-Guided MCMC Gaussian Allocation.*” We provide:

- A large-scene benchmark comparison (§2).
- Full SfM runtime tables for the Courthouse scene (§3).
- Full per-scene qualitative comparisons on all nine Mip-NeRF 360 scenes (§4).
- Per-image LPIPS analysis across ablation variants (§5).
- Threshold-sensitivity sweeps for τ_{imp} and the number of scoring views K (§6).
- Analysis of the SalientGS front-end: track-length distributions and a controlled track-cap ablation establishing that sufficient multi-view track coverage is a prerequisite for high reconstruction quality (§7).

2 Large-Scene Benchmark Comparison

Table 1 reports SSIM, PSNR, and LPIPS on Building, Rubble, Residence, and Sci-Art; each competing method is cited in the first column. Our method achieves the best PSNR on Building, Rubble, and Residence, the best LPIPS on Building, Residence, and Sci-Art, and the best SSIM on Rubble, indicating that the unified SfM-to-3DGS pipeline remains highly competitive even on challenging large-scale scenes. Under the same large-scene training setup, SalientGS averages only about 1 h wall-clock training and ~ 5 GB peak GPU memory, compared to about 3 h and ~ 14 GB for CityGS-X [5].

3 SfM Runtime Details

Table 2 reports the full SfM component runtime on the Courthouse scene with 250–1000 input images, including GloSplat-F (XFeat+LightGlue with retrieval top-5; see §7 for details). These numbers correspond to the trend summarized by Figure 5 in the main paper.

4 Per-Scene Qualitative Comparisons

We present qualitative comparisons on all nine Mip-NeRF 360 [1] scenes. Each figure shows the rendered output from SalientGS (Ours) alongside ablation variants (Vanilla MCMC, Standard ADC, Freeze Poses) and a baseline method, with zoomed crops at the region of maximum LPIPS difference.

Outdoor Scenes. Figures 1–5 cover the five outdoor scenes. SalientGS consistently resolves fine-grained texture—foliage, bark, grass—that Vanilla MCMC blurs and Standard ADC can render with color artifacts on most scenes. The advantage is most pronounced on *stump* and *flowers*, where high-frequency detail benefits most from importance-guided capacity allocation; on *garden*, SalientGS and Standard ADC perform comparably.

Indoor Scenes. Figures 6–9 cover the four indoor scenes. On *counter*, *bonsai*, and *kitchen*, SalientGS outperforms all ablation variants, though the margins are smaller than on outdoor scenes due to simpler geometry and more uniform textures. On *room*, SalientGS underperforms VGGT-X on flat, textureless surfaces where first-order SfM initialization provides weaker geometric constraints than learned pose estimators (see main paper, Qualitative Results and Conclusion), though it still leads all ablation variants in the per-image LPIPS table.

5 Per-Image LPIPS Analysis

Table 3 reports per-scene LPIPS (lower is better) for each Mip-NeRF 360 scene across SalientGS and three ablation variants. The scene rows are per-scene means, while the final row is the pooled mean over all rendered test images; it is therefore not an unweighted average of the nine scene rows. SalientGS achieves the best LPIPS on all 9 scenes, with its strongest absolute margins on outdoor scenes with complex textures (*bicycle*, *flowers*, *stump*, *treehill*). The indoor margins are smaller, consistent with the reduced benefit of importance-guided allocation on scenes with uniform textures, though SalientGS still leads its ablation variants on *bonsai*, *kitchen*, and *room*.

6 Threshold Sensitivity Sweep

Table 4 reports the Mip-NeRF 360 sensitivity sweep referenced in the main paper. We vary the importance threshold τ_{imp} and the number of views K used for multi-view score computation while keeping the rest of the pipeline fixed.

The default setting in the main paper is $\tau_{\text{imp}}=5$ and $K=10$, corresponding to the full model. PSNR and SSIM vary mildly across nearby settings, while LPIPS is more sensitive to perceptual blur and high-frequency detail; we therefore report

Table 1: Large-scene benchmark comparison. Metrics are SSIM $\uparrow$, PSNR $\uparrow$, and LPIPS $\downarrow$. Red / orange / yellow denote the best / second-best / third-best values in each column.

Method	Building			Rubble			Residence			Sci-Art		
	SSIM	PSNR	LPIPS	SSIM	PSNR	LPIPS	SSIM	PSNR	LPIPS	SSIM	PSNR	LPIPS
<i>w/ Geometric Optimization</i>												
SuGaR [6]	0.507	17.76	0.455	0.577	20.69	0.453	0.603	18.74	0.406	0.698	18.60	0.349
NeuS [15]	0.463	18.01	0.611	0.480	20.46	0.618	0.503	17.85	0.533	0.633	18.62	0.472
Neuralangelo [9]	0.582	17.89	0.322	0.625	20.18	0.314	0.644	18.03	0.263	0.769	19.10	0.231
PGSR [2]	0.480	16.12	0.573	0.728	23.09	0.334	0.746	20.57	0.289	0.799	19.72	0.275
VCR-Gaus [3]	0.502	19.56	0.502	0.541	21.34	0.428	0.623	20.59	0.359	0.665	19.31	0.465
CityGaussianV2 [12]	0.650	19.07	0.397	0.720	23.75	0.322	0.769	21.15	0.234	0.810	20.66	0.266
UrbanGS [8]	0.802	22.82	0.208	0.791	26.25	0.210	0.823	22.48	0.205	0.824	22.62	0.279
CityGS-X [5]	0.817	22.76	0.191	0.823	26.15	0.210	0.819	22.44	0.194	0.867	22.77	0.179
<i>w/o Geometric Optimization</i>												
Mega-NeRF [14]	0.547	20.92	0.454	0.553	24.06	0.508	0.628	22.08	0.401	0.770	25.60	0.312
Switch-NeRF [13]	0.579	21.54	0.397	0.562	24.31	0.478	0.654	22.57	0.352	0.795	26.51	0.271
VastGaussian [10]	0.728	21.80	0.225	0.742	25.20	0.264	0.699	21.01	0.261	0.761	22.64	0.261
3DGS [7]	0.738	22.53	0.214	0.725	25.51	0.316	0.745	22.36	0.247	0.791	24.13	0.262
DoGaussian [4]	0.759	22.73	0.204	0.765	25.78	0.257	0.740	21.94	0.244	0.804	24.42	0.219
CityGaussian [11]	0.778	21.55	0.246	0.813	25.77	0.228	0.813	22.00	0.211	0.837	21.39	0.230
SalientGS (Ours)	0.804	24.19	0.181	0.824	26.98	0.213	0.816	22.67	0.193	0.857	22.91	0.173

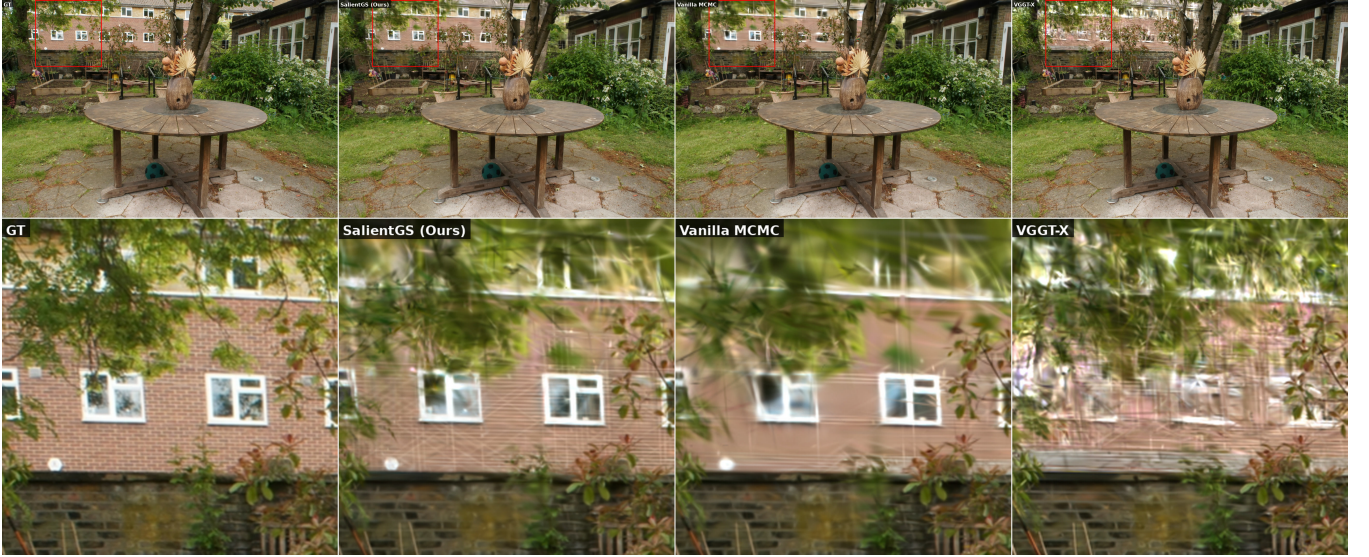
**Figure 1: Per-scene qualitative comparison: *bicycle*.**

the three metrics separately rather than summarizing robustness with a single PSNR range.

7 Front-End Track Analysis and Track-Length Ablation

A natural question is whether the quality advantage of SalientGS over retrieval-only front-ends (such as GloSplat-F, which

uses XFeat+LightGlue with top-5 retrieval) originates primarily from better Gaussian allocation or from a richer multi-view track structure. We investigate this through two complementary experiments: (1) a comparison of the 3D-point track-length distributions produced by three front-ends on the *bicycle* scene, and (2) a controlled track-length cap ablation that holds the front-end fixed and varies only the

Figure 2: Per-scene qualitative comparison: *flowers*.Figure 3: Per-scene qualitative comparison: *garden*.

maximum number of observations retained per track before global positioning and bundle adjustment.

Track-Length Distributions. Table 5 compares the track-length distributions produced on the `bicycle` scene by three pipelines: GloSplat-A (SIFT exhaustive), GloSplat-F (XFeat+LightGlue retrieval top-5), and SalientGS (SIFT + Fisher Vector retrieval, $k=20$). SalientGS reconstructs $\sim 4\%$ fewer 3D points than GloSplat-A (51.8k vs. 54.3k), but its track-length distribution remains very close: median and max are nearly identical, the mean is only marginally lower (4.47 vs. 4.69), and all four length-bucket percentages differ by at most 0.6 pp. In

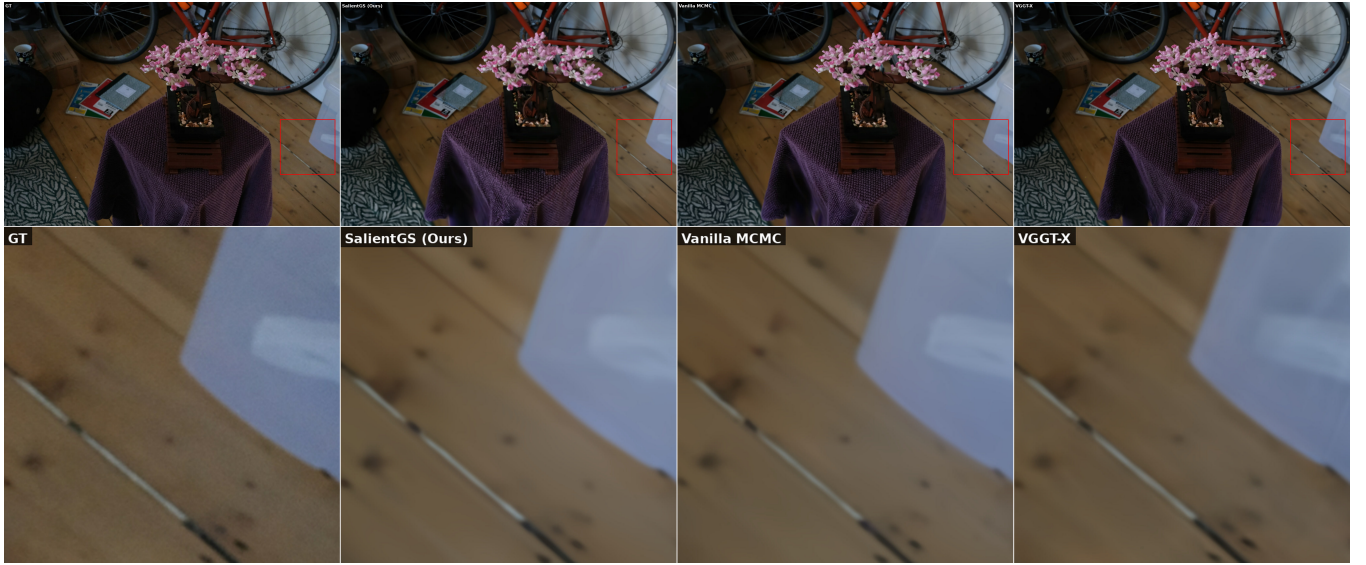
contrast, GloSplat-F—which uses only 5 retrieval neighbors—yields $\sim 16\%$ fewer points with a substantially shorter track distribution (median 3 vs. 4, max 20 vs. 74), and long tracks (≥ 9 observations) are nearly absent (2.1% vs. 9.5%). The MST connectivity guarantee built into SalientGS’s Fisher Vector retrieval (Sec. 3.1 of the main paper) is responsible for preserving the exhaustive-like track structure: the $k=20$ candidate graph, augmented with MST edges, achieves far denser coverage than a bare top-5 list.

Track-Length Cap Ablation. To establish causality between track length and downstream rendering quality, we conduct a

Figure 4: Per-scene qualitative comparison: *stump*.Figure 5: Per-scene qualitative comparison: *treehill*.

controlled ablation: we fix each method’s front-end (i.e., the same pair graph and matches) and vary only the maximum number of observations retained per track before global positioning and bundle adjustment. Table 6 reports the effect on Mip-NeRF 360 (averaged over 9 scenes). The $L=2$ condition reduces every track to a pairwise observation—simulating a worst-case version of the track shortening seen in GloSplat-F—while $L=all$ retains the full distribution. For GloSplat (SIFT exhaustive front-end), capping at $L=2$ causes a 1.40 dB PSNR drop ($28.86 \rightarrow 27.46$), confirming that long multi-view tracks provide substantial geometric constraints that improve triangulation robustness and stabilize bundle adjustment. SalientGS

exhibits a nearly identical degradation of 1.36 dB ($28.89 \rightarrow 27.53$) under the same $L=2$ cap. This convergence is expected: once every track is reduced to a pairwise observation, both pipelines operate on geometrically equivalent constraints regardless of their front-end or allocation strategy, leaving the underlying bundle adjustment quality as the dominant factor. The full-track advantage ($L=all$) therefore reflects a shared structural property of multi-view triangulation rather than any method-specific component, further reinforcing that sufficient track length is a prerequisite for high reconstruction quality independent of the downstream renderer.

Figure 6: Per-scene qualitative comparison: *bonsai*.Figure 7: Per-scene qualitative comparison: *counter*.

Taken together, these results support two conclusions. First, SalientGS’s Fisher Vector retrieval with $k=20$ and MST connectivity is sufficient to recover an exhaustive-quality track distribution at a fraction of the matching cost, closing the structural gap that limits GloSplat-F. Second, the near-identical $L=2$ scores across both methods confirm that the gain from full tracks is a property of multi-view triangulation geometry, not of any particular front-end or allocation design—making rich track coverage a necessary condition for state-of-the-art reconstruction quality.

8 Additional Camera-Ready Ablations

This section records the additional measured experiments completed during the author response. The results are included to make the source-of-gain decomposition, schedule choices, and fixed-budget behavior explicit.

The controlled matrix in Table 7 shows that guided allocation improves both initializers (LPIPS $0.151 \rightarrow 0.133$ for GLOMAP and $0.149 \rightarrow 0.131$ for FastMap). Thus the main improvement is attributable to importance-space allocation rather than simply to a stronger camera initializer.

Figure 8: Per-scene qualitative comparison: *kitchen*.Figure 9: Per-scene qualitative comparison: *room*. SalientGS underperforms on this indoor scene with uniform textures, where first-order SfM provides weaker geometric constraints.

9 Additional Robustness and Failure Diagnostics

The diagnostics in Table 15 identify the texture-poor *room* case through collapsed track length, greater MST reliance, and a high early residual. A feed-forward fallback improves the failure-view LPIPS from 0.312 to 0.235, slightly better than the VGGT-X reference of 0.244. Sparse/few-view reconstruction and severe repetitive or symmetric appearance remain limitations; learned retrieval and adaptive geometry-aware budgets are natural future extensions.

Table 2: SfM runtime comparison (seconds). We compare the SfM component runtime across different methods on the Courthouse scene with varying numbers of input images. SalientGS combines Fisher Vector retrieval with first-order SfM optimization to achieve significant speedups over traditional and learning-based approaches.

# Images	COLMAP	MASt3R	GloSplat-A	VGGT-X	GloSplat-F	SalientGS
250	644.6	312.2	222.2	53.9	100.2	47.8
500	1165.0	638.0	408.2	113.3	191.9	98.0
750	2226.3	978.5	855.7	249.6	250.0	138.5
1000	4349.2	1308.2	1371.3	398.9	327.8	186.1

Table 3: Mean per-image LPIPS ($\downarrow$) on Mip-NeRF 360 scenes. Scene rows show per-scene means; the final row is the pooled test-image mean. Best result per scene in bold.

Scene	SalientGS	Vanilla MCMC	Std. ADC	Freeze Poses
<i>Outdoor Scenes</i>				
bicycle	0.088	0.152	0.153	0.173
flowers	0.201	0.264	0.309	0.293
garden	0.018	0.074	0.066	0.081
stump	0.075	0.140	0.145	0.151
treehill	0.186	0.260	0.269	0.270
<i>Indoor Scenes</i>				
bonsai	0.143	0.169	0.171	0.170
counter	0.079	0.132	0.148	0.134
kitchen	0.155	0.236	0.167	0.173
room	0.230	0.271	0.252	0.271
Average	0.131	0.189	0.187	0.191

Table 4: Threshold-sensitivity sweep on Mip-NeRF 360. We vary the importance threshold τ_{imp} and the number of scoring views K . The default configuration used in the main paper is marked as *default*.

Group	Value	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
τ_{imp}	1	28.688	0.8444	0.159
	3	28.685	0.8448	0.157
	5 (<i>default</i>)	28.890	0.8570	0.131
	10	28.732	0.8466	0.153
	20	28.759	0.8477	0.151
K	3	28.696	0.8457	0.156
	5	28.689	0.8455	0.156
	10 (<i>default</i>)	28.890	0.8570	0.131
	15	28.714	0.8450	0.157
	20	28.679	0.8446	0.157

Table 5: Track-length distribution on the bicycle scene for GloSplat-A (SIFT exhaustive), GloSplat-F (XFeat+LightGlue, retrieval top-5), and SalientGS (Fisher Vector, $k=20$).

	GloSplat-A	GloSplat-F	SalientGS
Total points	54,275	45,507	51,847
Mean / Median	4.69 / 4	3.75 / 3	4.47 / 4
Max	74	20	71
2–3	26,861 (49.5%)	27,182 (59.7%)	25,814 (49.8%)
4–8	22,036 (40.6%)	17,366 (38.2%)	21,124 (40.7%)
9–16	4,694 (8.6%)	948 (2.1%)	4,133 (8.0%)
17+	684 (1.3%)	11 (0.0%)	776 (1.5%)

Table 6: Track-length cap ablation on MipNeRF360 (9-scene avg.). The front-end is fixed for each method; only the maximum retained observations per track varies.

Max track length	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
<i>GloSplat (SIFT exhaustive front-end)</i>			
<i>GloSplat-F (retrieval top-5, ref.)</i>	27.77	0.818	0.164
$L = 2$	27.46	0.804	0.174
$L = \text{all}$	28.86	0.862	0.139
<i>GloSplat-A (exhaustive, ref.)</i>	28.86	0.862	0.139
<i>SalientGS (Fisher Vector $k=20$ front-end)</i>			
$L = 2$	27.53	0.808	0.172
$L = \text{all}$	28.89	0.857	0.131
<i>SalientGS (full model, ref.)</i>	28.89	0.857	0.131

Table 7: SfM initialization versus Gaussian allocator on Mip-NeRF 360. All rows use the same joint training protocol.

SfM initialization	Allocator	PSNR↑	SSIM↑	LPIPS↓
GLOMAP	ADC	27.52	0.827	0.168
GLOMAP	Vanilla MCMC	28.70	0.846	0.151
GLOMAP	Guided MCMC	28.87	0.855	0.133
FastMap	ADC	27.67	0.828	0.162
FastMap	Vanilla MCMC	28.72	0.848	0.149
FastMap	Guided MCMC (ours)	28.89	0.857	0.131

Table 8: Importance-score schedule and opacity mixing on Mip-NeRF 360.

Setting	PSNR↑	SSIM↑	LPIPS↓
$T = 100$	28.83	0.855	0.135
$T = 200$	28.87	0.856	0.133
$T = 500$ (default)	28.89	0.857	0.131
$T = 1000$	28.80	0.852	0.137
score at iter. 0	28.78	0.851	0.138
score at iter. 3000	28.89	0.857	0.131
$\lambda_{\text{mix}} = 0$	28.78	0.851	0.141
$\lambda_{\text{mix}} = 0.05$	28.89	0.857	0.131
$\lambda_{\text{mix}} = 0.20$	28.75	0.850	0.145

Table 9: Robust quantile normalization and selection-quantile interaction on Mip-NeRF 360. The final row reports the fraction of Gaussians in the high-error mass at 30K iterations.

Setting	PSNR↑	SSIM↑	LPIPS↓
Plain min-max	28.71	0.850	0.150
Default quantiles	28.89	0.857	0.131
$(q_{\text{hi}}, q_{\text{lo}}) = (.80, .20)$	28.79	0.852	0.139
$(q_{\text{hi}}, q_{\text{lo}}) = (.95, .05)$	28.82	0.851	0.137
High-error mass at 30K	robust: 9.1%; min-max: 42.5%		

Table 10: Pose error before and after joint optimization on the ETH3D SLAM evaluation. Errors are in centimeters; percentages are fractions of cameras below the indicated threshold.

Method	Median↓	Mean↓	< 5cm↑	< 10cm↑
FastMap init	9.8	18.5	31.6	50.5
FastMap + joint	6.9	13.5	38.2	57.0
GLOMAP init	8.5	14.6	35.5	55.5
GLOMAP + joint	7.1	12.6	37.8	56.5

Table 11: Per-scene Deep Blending results. The competing methods fail on *drjohnson*, which explains the averaged values in the main table.

Scene	Method	PSNR↑	SSIM↑	LPIPS↓
<i>playroom</i>	GloSplat-A	29.38	0.903	0.224
<i>playroom</i>	SalientGS	30.36	0.926	0.140
<i>drjohnson</i>	GloSplat-A	7.52	0.263	0.792
<i>drjohnson</i>	VGGT-X	7.65	0.337	0.858
<i>drjohnson</i>	SalientGS	29.62	0.916	0.148

Table 12: Budget efficiency on Mip-NeRF 360.

N_{GS}	Vanilla MCMC	Guided MCMC	ΔPSNR
500K	28.19	28.46	+0.27
1.0M	28.59	28.81	+0.22
1.5M	28.72	28.89	+0.17
2.0M	28.80	28.93	+0.13
3.0M	28.88	28.97	+0.09

Table 13: Dense indoor robustness. SG denotes SalientGS and V+M denotes the VGGT-X+MCMC comparison.

Scene	SG PSNR	V+M PSNR	SG LPIPS	V+M LPIPS
bonsai	33.58	30.71	0.105	0.139
counter	29.59	28.80	0.137	0.158
kitchen	32.91	29.68	0.077	0.113
room	32.04	30.76	0.149	0.165
Average	32.03	29.99	0.117	0.144

Table 14: Additional large-scene and drone-capture evidence.

Setting	Result
Mill-19 Building	Best PSNR, 24.19 dB
Mill-19 Rubble	Best PSNR, 26.98 dB
Four large scenes	Best PSNR/LPIPS on 3 of 4 scenes
Runtime/memory	~1 h / ~5 GB vs. ~3 h / ~14 GB for CityGS-X

Table 15: Failure diagnostics and fallback on representative Mip-NeRF 360 scenes. Residual is the normalized early epipolar residual.

Scene	Median track	Mean degree	MST reliance	Residual
garden	4	18.4	4%	0.20
kitchen	3	13.4	11%	0.43
room	2	9.7	21%	0.57

room LPIPS: original 0.312; fallback 0.235; VGGT-X 0.244

Table 16: Sequential Tanks & Temples comparison with CF-3DGS.

Scene	Method	PSNR↑	SSIM↑	LPIPS↓
Family	CF-3DGS	33.12	0.960	0.051
Family	SalientGS	36.80	0.970	0.024
Francis	CF-3DGS	32.78	0.915	0.135
Francis	SalientGS	36.29	0.947	0.056

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